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## Heating cooking apparatus

### Abstract


A heating cooking apparatus includes a main body, a door, a latch head provided at the door, a holding member provided at the main body and that holds the latch head to cause the door to be in a locked state, a door opening button, a door unlock key, a displacement transmission mechanism that transmits a displacement of the door opening button to cause that displacement to release the locked state of the door by the holding member, a locking cam that is movable between a limiting position at which at least one of the displacement of the door opening button and the transmission by the displacement transmission mechanism is limited, and a non-limiting position at which neither is limited, and a driving unit that causes the locking cam to move to the non-limiting position in response to an operation of the door unlock key.



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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims the benefit of priority to Japanese Patent Application Number 2022-123292 filed on Aug. 2, 2022. The entire contents of the above-identified application are hereby incorporated by reference.

BACKGROUND

Technical Field

(2) The present disclosure relates to a heating cooking apparatus such as a microwave oven that heats an object to be heated by using microwaves or the like. More particularly, the present disclosure relates to a heating cooking apparatus having a configuration in which a door of a front face opening of a heating chamber is opened by a mechanical operation performed on a push button, a handle, or the like.

(3) In the heating device disclosed in JP 2010-139120 A, a hook 5 is inserted, intentionally by a

user or automatically, into an insertion hole 6 in an upper surface of an opening/closing door (door) 3 of a heating chamber 2 included in a heating device main body 1 so that the door 3 does not open during heating or when the temperature of the heating chamber is high. With this configuration, the heating device is configured such that the opening of the door 3 is restricted to prevent an unintended operation of the door 3.

SUMMARY

(4) To open the door 3 in the heating device described in JP 2010-139120 A, it is necessary to perform the following two-stage mechanical operation. (1) Unlock the lock formed by the hook 5 by opening an operation unit front plate 10 and sliding a knob 7 upward. (2) Grip a handle 4 and open the door 3 toward the front.

(5) The second mechanical operation of the two operations has been always required in many heating devices in the related art, but the heating device described in JP 2010-139120 A also requires the first mechanical operation. In other words, since a two-stage mechanical operation is required every time the door 3 is to be opened, some users may feel that such an operation is troublesome.

(6) Further, whenever the user performs the mechanical operations for locking the door 3 or unlocking the lock of the door 3, for example, the user may feel it inconvenient that the operations require both hands in some cases, or the user may forget to perform the operation for locking the door 3 because the operation is troublesome.

(7) An object of the present disclosure is to provide a heating cooking apparatus with which it is possible, with one hand, to easily perform a non-mechanical operation (first mechanical operation described above) to unlock a lock before opening a door, and to also automatically lock the door.

(8) A heating cooking apparatus according to the present disclosure includes a main body including a heating chamber accommodating an object to be heated, a door configured to open and close a front face opening of the heating chamber, a locking member provided at the door, a holding member provided at the main body and configured to hold the locking member to cause the door to be in a locked state, a first operation member configured to be displaced between a first position and a second position and to be operated to open the door, a second operation member configured to be operated to release the locked state of the door, a displacement transmission mechanism configured to transmit a displacement of the first operation member from the first position, and to release the locked state of the door by the holding member by causing the first operation member to be displaced to the second position, a displacement transmission limiting member configured to be movable between a limiting position at which at least one of the displacement of the first operation member and the transmission by the displacement transmission mechanism is limited, and a non-limiting position at which neither the displacement of the first operation member nor the transmission by the displacement transmission mechanism is limited, and a driving unit configured to cause the displacement transmission limiting member to move to the non-limiting position in response to an operation of the second operation member.

(9) According to the present disclosure, it is possible, with one hand, to easily perform a non-mechanical operation to release a lock before opening a door of a heating cooking apparatus, and to also automatically lock the door.

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## Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The disclosure will be described with reference to the accompanying drawings, wherein like numbers reference like elements.

(2) FIG. **1** is a front view of a heating cooking apparatus **100** according to a first embodiment of the present disclosure with a door **12** closed.

(3) FIG. **2** is a front view of the heating cooking apparatus **100** with the door **12** open.
(4) FIG. **3** is a schematic view illustrating an example of an operation panel **16** of the heating cooking apparatus **100**.
(5) FIG. **4**A is a perspective view of a locking unit **30** disposed at or near a door opening button **15** inside a main body **10**, and the vicinity of the locking unit **30**.
(6) FIG. **4**B is a cross-sectional view for explaining a basic operation of opening the door **12** by a pressing operation on the door opening button **15**.
(7) FIG. **5**A is a perspective view illustrating a rotational position of a locking cam **31** inside the locking unit **30** in a door locked state.
(8) FIG. **5**B is a cross-sectional view in the door locked state.
(9) FIG. **6**A is a perspective view illustrating a rotational position of the locking cam **31** inside the locking unit **30** in a door unlocked state.
(10) FIG. **6**B is a cross-sectional view in the door unlocked state.
(11) FIG. **7** is a block diagram illustrating an overall electrical configuration of the heating cooking apparatus **100**.
(12) FIG. **8** is a flowchart illustrating processing performed by a control unit **40** when a door unlock key **16***b* is operated.
(13) FIG. **9** is a flowchart illustrating processing performed by the control unit **40** when a heating start key **16***d* is operated.
(14) FIG. **10** is a flowchart illustrating processing performed by the control unit **40** when the door **12** is opened within an extremely short time period after the door unlock key **16***b* has been operated.
(15) FIG. **11** is a perspective view illustrating a rotational position of the locking cam **31** in the door locked state of a locking unit **30**A according to a first modified example of the first embodiment of the present disclosure.
(16) FIG. **12**A is a plan view illustrating a rotational position of the locking cam **31** in the door locked state of the locking unit **30**A.
(17) FIG. **12**B is a plan view illustrating a rotational position of the locking cam **31** in the door unlocked state of the locking unit **30**A.
(18) FIG. **13**A is a plan view illustrating a rotational position of a locking cam **31**B in the door locked state of a locking unit **30**B according to a second modified example of the first embodiment of the present disclosure.
(19) FIG. **13**B is a plan view illustrating a rotational position of the locking cam **31**B in the door unlocked state of the locking unit **30**B.
(20) FIG. **14** is a perspective view of the locking cam **31**B as viewed from below.

DESCRIPTION OF EMBODIMENTS

(21) Embodiments of the present disclosure will be described with reference to the drawings. Note that, in the drawings, the same or equivalent components are denoted by the same reference numerals and signs, and description thereof will not be repeated.

First Embodiment

1.1 Overall Basic Configuration

(22) First, a basic configuration of a first embodiment of the present disclosure will be described. FIG. **1** is a front view of a heating cooking apparatus **100** according to the first embodiment of the present disclosure with a door **12** closed. FIG. **2** is a front view of the heating cooking apparatus **100** with the door **12** open. FIG. **3** is a schematic view illustrating an operation panel **16** of the heating cooking apparatus **100**.
(23) As illustrated in FIGS. **1** and **2**, the heating cooking apparatus **100** includes a box-like main body **10** including a heating chamber **11** that accommodates an object to be heated, the door **12** that opens and closes a front face opening **11***a* of the main body **10**, a door opening button **15**, and the operation panel **16**.

(24) The door **12** is pivotally supported on the left side of the front face opening **11***a*. A heat-resistant glass window **13** is disposed at a central portion of the door **12** so that the inside of the heating chamber **11** can be seen from outside, and latch heads **14** are provided at two locations on the back side of a side end portion, of the door **12**, that is closer to the operation panel **16**. Interlock mechanisms **17** for holding the latch heads **14** are provided at corresponding positions on the main body **10** side. With the interlock mechanisms **17**, the heating cooking apparatus **100** is configured so that a heating operation in which microwaves are emitted is not performed unless the door **12** is securely closed.
(25) On the right side of the front face opening **11***a*, the door opening button **15** having a horizontally long rectangular shape is provided at or near the lower end of the front face opening **11***a*, and the operation panel **16** having a vertically long rectangular shape is disposed in a remaining portion of the right side of the front face opening **11***a*.
(26) The door opening button **15** is configured so that a door locked state is released and the door **12** opens when a pressing operation is performed on the door opening button **15** by a user.
(27) As illustrated in FIG. **3**, the operation panel **16** includes a display **16***a*, a door unlock key **16***b*, a door locked state indicator **16***c*, a heating start key **16***d*, and a stop/cancel key **16***e*.
(28) The display **16***a* is disposed at or near an upper edge of the operation panel **16**, and can display four numbers or characters. The display **16***a* displays a set heating time, a remaining time during heating, a heating output level, a menu number, and the like.
(29) The door unlock key **16***b* is an operation key that needs to be pressed immediately before the pressing operation on the door opening button **15**, and is disposed on the right side immediately below the display **16***a*. When the door unlock key **16***b* is pressed, the door locked state is released. Then, the door **12** is opened by a subsequent pressing operation on the door opening button **15**. This heating cooking apparatus **100** is configured so that unless the door unlock key **16***b* is pressed, the door locked state is not released and thus the door opening button **15** cannot be pressed. Further, the door locked state indicator **16***c*, which allows the user to visually confirm whether the door is in the locked state, is disposed on the left side of the door unlock key **16***b*. In the door locked state indicator **16***c*, a character string “door locked” is lit up brightly when the door is in the locked state, and the character string is not lit up and disappears when the door is in a door unlocked state. Note that the door locked state indicator **16***c* corresponds to a “display unit” of the present disclosure.
(30) The heating start key **16***d* is an operation key for starting the heating operation. The stop/cancel key **16***e* is an operation key for stopping the heating operation or canceling various settings during the heating operation. Note that the operation panel **16** also includes various other keys, but description thereof is omitted herein.
1.2 Locking Unit **30** Disposed Inside Main Body **10**
(31) FIG. **4**A is a perspective view of the locking unit **30** disposed at or near the door opening button **15** inside the main body **10**, and the vicinity of the locking unit **30**. FIG. **4**B is a cross-sectional view for explaining a basic operation of opening the door **12** by performing the pressing operation on the door opening button **15**. FIG. **5**A is a perspective view illustrating a rotational position PL of a locking cam **31** inside the locking unit **30** in the door locked state. FIG. **5**B is a cross-sectional view in the door locked state. FIG. **6**A is a perspective view illustrating a rotational position PNL of the locking cam **31** inside the locking unit **30** in the door unlocked state. FIG. **6**B is a cross-sectional view in the door unlocked state. Note that, in FIGS. **5**A and **6**A, some members constituting the locking unit **30** are not illustrated to make it easier to see the rotational positions of the locking cam **31**.
(32) As illustrated in these drawings, the locking unit **30** includes the locking cam **31** relating to limitation of the basic operation of opening the door **12**, a cam phase detection switch **32** that detects the rotational position of the locking cam **31**, and a synchronous motor **33** that rotates the locking cam **31**.

(33) The locking cam **31** has a shape having columnar outer peripheral surfaces at both ends, and is rotationally movable at least between the rotational position PL at which the basic operation of opening the door **12** is limited and the rotational position PNL at which the basic operation is not limited. Note that the rotational position PL and the rotational position PNL correspond to a “limiting position” and a “non-limiting position” of the present disclosure, respectively. When opening the door **12**, the locking cam **31** is moved to the rotational position PNL so as not to interfere with the pressing operation on the door opening button **15**. Details of this operation will be described below with reference to FIGS. **6**A and **6**B.
(34) The cam phase detection switch **32** detects the phase corresponding to the rotational position of the locking cam **31**, as ON or OFF. In this locking cam **31**, a rotational position at which the cam phase detection switch **32** changes from ON to OFF is set as an initial position corresponding to the door locked state. Examples of the cam phase detection switch **32** include, but are not limited to, a micro switch.
(35) The synchronous motor **33** rotates the locking cam **31** in one direction, for example, the counterclockwise (CCW) direction. The motor for rotating the locking cam **31** is not limited to the synchronous motor, and may be a stepper motor, for example. Further, the rotational direction is not limited to one direction and may be both directions. However, it is assumed that a switch or the like necessary for accurately detecting the rotational position of the locking cam **31** (at least the position corresponding to the door locked state) is provided.
(36) In addition, the heating cooking apparatus **100** includes a first lever **22** and a second lever **23** inside the main body **10**, which transmit a displacement caused by the pressing operation performed on the door opening button **15**. The displacement of the door opening button **15** is transmitted from the second lever **23** to the first lever **22**, and further to the latch head **14** that engages with a latch hook **21**.
(37) When the pressing operation on the door opening button **15** is performed, an inner portion of the door opening button **15** comes into contact with the second lever **23** and rotates the second lever **23**. As a result, the second lever **23** comes into contact with the first lever **22** and pushes up the first lever **22**. Then, the pushed up first lever **22** comes into contact with the latch head **14** provided on the back side of the door **12** and pushes up the latch head **14**. As a result, the engagement between the latch head **14** and the latch hook **21** is released, and the door **12** is opened.
(38) As illustrated in FIGS. **5**A and **5**B, when the locking cam **31** of the locking unit **30** is moved to the rotational position PL at which the basic operation for opening the door **12** is limited, the outer peripheral surface of the locking cam **31** is in close proximity to the inner portion of the door opening button **15**.
(39) Thus, even if an attempt is made to perform the pressing operation on the door opening button **15**, the inner portion of the door opening button **15** immediately comes into contact with the locking cam **31**, and the door opening button cannot be pressed in any further. As a result, since the second lever **23** does not rotate, the first lever **22** is not pushed up, and thus the latch head **14** is not pushed up either, so the engagement between the latch head **14** and the latch hook **21** is not released. In other words, a state (door locked state) is achieved in which the door **12** cannot be opened by performing the pressing operation on the door opening button **15**.
(40) On the other hand, as illustrated in FIGS. **6**A and **6**B, when the locking cam **31** of the locking unit **30** is moved to the rotational position PNL, the outer peripheral surface of the locking cam **31** is sufficiently separated from the inner portion of the door opening button **15**.
(41) Thus, when the pressing operation on the door opening button **15** is performed, the basic operation of opening the door **12**, which is described above with reference to FIGS. **4**A and **4**B, is performed. In other words, a state (door unlocked state) is achieved in which the door **12** can be opened by performing the pressing operation on the door opening button **15**.
(42) Note that the locking unit **30** is an example of a main configuration for realizing the heating cooking apparatus of the present disclosure. Since it is sufficient simply to dispose and fix the

locking unit **30** at or near the door opening button **15** inside the main body **10**, changing the actual design of the main body **10** or the like is not necessary. Thus, the locking unit **30** can be retrofitted to many existing heating cooking apparatuses.

1.3 Electrical Configuration

(43) FIG. **7** is a block diagram illustrating an overall electrical configuration of the heating cooking apparatus **100**.

(44) As illustrated in FIG. **7**, the heating cooking apparatus **100** includes a control unit **40** that performs overall control.

(45) A door state switch **41** that detects an open/closed state of the door **12** and the cam phase detection switch **32** that detects the phase corresponding to the rotational position of the locking cam **31** are connected to an input to the control unit **40**.

(46) On the other hand, the operation panel **16**, an alarm **44**, a motor drive circuit **42** that drives the synchronous motor **33**, and a magnetron drive circuit **43** that drives a magnetron (not illustrated) for emitting microwaves are connected to an output from the control unit **40**. Note that the alarm **44** corresponds to a “notification unit” of the present disclosure.

1.4 Flowchart of Main Processing

(47) FIG. **8** is a flowchart illustrating processing performed by the control unit **40** when the door unlock key **16***b* is operated.

(48) The control unit **40** monitors whether an operation is performed on each of the keys disposed on the operation panel **16**, and controls the heating cooking apparatus **100** according to the key to which an operation has been performed. The control unit **40** also monitors whether an operation has been performed on the door unlock key **16***b* disposed at the operation panel **16**. Note that, although it is assumed that the door **12** is in the closed state at the start of processing described below, a condition determination such as step S**22** in FIG. **9** described below may be added. Alternatively, monitoring the presence and absence of an operation on each of the keys disposed in the operation panel **16** may be performed only when the door **12** is closed.

(49) As illustrated in FIG. **8**, at step S**11**, the control unit **40** checks whether the door unlock key **16***b* has been operated, and if no operation has been performed, the processing returns to the original state. If an operation has been performed, the processing proceeds to the next step S**12**.

(50) At step S**12**, the phase of the locking cam **31** is initialized. Specifically, the synchronous motor **33** is driven by the motor drive circuit **42**, and when it is detected that the cam phase detection switch **32** has changed from OFF to ON and then changed to OFF once again, the synchronous motor **33** is braked and stopped. As a result, the locking cam **31** stops at the initial position corresponding to the door locked state (see FIGS. **5**A and **5**B).

(51) At step S**13**, the synchronous motor **33** is driven by the motor drive circuit **42** for a predetermined time period (430 ms in this example) to rotate the locking cam **31** by 90 degrees. As a result, the locking cam **31** is moved to the rotational position corresponding to the door unlocked state (see FIGS. **6**A and **6**B). Unlike when the locking cam **31** is rotationally moved to the initial position (step S**12**), the rotation by 90 degrees can be controlled only by controlling the driving time of the synchronous motor **33**. However, a specific driving time needs to be determined in advance based on specifications of the synchronous motor **33** or the like.

(52) At step S**14**, an alarm is sounded twice to notify the user that the door has entered the door unlocked state. However, this step may be omitted.

(53) Thereafter, at step S**15**, the state of the door state switch **41** is monitored, and if the door **12** is changed to the open state within 10 seconds, the series of processing is terminated. However, if the state of the door state switch **41** has not changed, it is determined that the user has attempted to open the door **12** but has not actually opened the door **12**, and the processing proceeds to step S**16**.

(54) Since it is not preferable for the door **12** to be left in the door unlocked state even if the door **12** is closed, the door **12** is returned to the door locked state once again. Prior to this, at step S**16**, the alarm is sounded once to notify the user. Here, changing the number of times the alarm is

sounded can clearly distinguish this state from when the door **12** has entered door unlocked state (step S**14**). However, this step may be omitted.
(55) At step S**17**, in the same manner as at step S**12**, the phase of the locking cam **31** is initialized. In other words, the locking cam **31** is moved to the initial position corresponding to the door locked state.
(56) FIG. **9** is a flowchart illustrating processing performed by the control unit **40** when the heating start key **16***d* is operated.
(57) As illustrated in FIG. **9**, at step S**21**, the control unit **40** checks whether the heating start key **16***d* has been operated, and if no operation has been performed, the processing returns to the original state. If an operation has been performed, the processing proceeds to the next step S**22**.
(58) At step S**22**, the state of the door state switch **41** is detected, and when the detection result indicates that the door **12** is in the closed state, the processing proceeds to the next step S**23**. Otherwise (when the detection result indicates that the door **12** is in the open state), the processing returns to the original state without performing the processing subsequent to step S**23**. This is because the heating operation in which microwaves are emitted is not performed unless the door **12** is securely closed, due to the interlock mechanisms **17** (see FIG. **2**).
(59) At step S**23**, the cam phase detection switch **32** detects the phase of the locking cam **31**. At step S**24**, the synchronous motor **33** is driven by the motor drive circuit **42** to start the rotational movement of the locking cam **31**. At step S**25**, when the control unit **40** detects that the cam phase detection switch **32** has changed from ON to OFF, the synchronous motor **33** is braked and stopped. As a result, the locking cam **31** stops at the initial position corresponding to the door locked state. Note that steps S**23** to S**25** are substantially the same as the processing at step S**12**.
(60) By the processing up to this point, it is confirmed that the door **12** is in the closed state, and that the door **12** has entered the door locked state in which the door **12** cannot be opened by performing the pressing operation on the door opening button **15**. Then, at step S**26**, the magnetron drive circuit **43** drives the magnetron to start emitting microwaves to heat an object to be heated.
(61) FIG. **10** is a flowchart illustrating processing performed by the control unit **40** when the door **12** is opened within an extremely short time period after the door unlock key **16***b* has been operated. Note that, in a normal case in which the door **12** is opened after the above-described short time period has elapsed from when the door unlock key **16***b* is operated, the processing according to the flowchart illustrated in FIG. **8** is performed.
(62) At step S**31**, in the same manner as at step S**11** in FIG. **8**, the control unit **40** checks whether the door unlock key **16***b* has been operated, and if it has been operated, the processing proceeds to the next step S**32**.
(63) At step S**32**, the state of the door state switch **41** is checked, and the processing proceeds to the next step S**33** only when the door **12** is changed to the open state within an extremely short time period (within 200 ms, for example). Otherwise, the processing proceeds to step S**34**.
(64) At step S**33**, the phase of the locking cam **31** is initialized in the same manner as at step S**12** in FIG. **8**. This is because, when the door opening button **15** is pressed within an extremely short time period after the door unlock key **16***b* has been operated, the locking cam **31** may be forcibly rotated by the door opening button **15**. If the locking cam **31** is forcibly rotated by an external force, the phase (rotational position) of the locking cam **31** cannot be correctly controlled. Therefore, the phase of the locking cam **31** needs to be initialized.
(65) At step S**34**, in the same manner as at step S**17** in FIG. **8**, the locking cam **31** is stopped at the initial position corresponding to the door locked state within a predetermined time period.
(66) At step S**35**, in the same manner as at step S**21** in FIG. **9**, the control unit **40** checks whether the heating start key **16***d* has been operated, and only when it has been operated, the processing proceeds to the next step **36**.
(67) At step S**36**, in the same manner as at step S**25** in FIG. **9**, the control unit **40** checks whether the locking cam **31** has been moved to the initial position corresponding to the door locked state,

and this check is repeated until the movement is completed. If the control unit **40** confirms that the movement has been completed, the processing proceeds to the next step S**37**.
(68) At step S**37**, in the same manner as at step S**26** in FIG. **9**, the magnetron is driven by the magnetron drive circuit **43** to start emitting microwaves to heat the object to be heated.
(69) Note that, to prevent this slightly unique situation from occurring as much as possible, it is preferable that the door unlock key **16***b* be disposed as far away from the door opening button **15** as possible. This is to ensure that a certain amount of time is required between operating the door unlock key **16***b* to performing the pressing operation on the door opening button **15**. For example, it is preferable that the door unlock key **16***b* be disposed at an uppermost portion of the operation panel **16**, or the door unlock key **16***b* be separated from the door opening button **15** by at least one half or more of the length of the operation panel **16** in the longitudinal direction (height direction) of the operation panel **16**.
First Modified Example of First Embodiment
(70) A configuration in which the locking unit **30** of the above-described first embodiment is replaced with a locking unit **30**A will be described below as a first modified example. In this locking unit **30**A, automatic reverse rotation of a synchronous motor and phase detection of a locking cam by using a micro switch similar to that of the first embodiment are employed.
(71) FIG. **11** is a perspective view illustrating a rotational position of the locking cam **31** in the door locked state of the locking unit **30**A according to the first modified example of the first embodiment of the present disclosure. FIG. **12**A is a plan view illustrating the rotational position of the locking cam **31** in the door locked state of the locking unit **30**A. FIG. **12**B is a plan view illustrating a rotational position of the locking cam **31** in the door unlocked state of the locking unit **30**A. Note that in these drawings, to make it easier to see the rotational positions of the locking cam **31**A, illustration of the synchronous motor and the like constituting the locking unit **30**A is omitted.
(72) The locking unit **30**A includes the locking cam **31**, a cam phase detection switch **32**A, the synchronous motor **33**, and a stopper pin **34**A.
(73) The synchronous motor **33** is driven by the motor drive circuit **42**. When the synchronous motor **33** is stopped at a timing at which an actuator of the cam phase detection switch **32**A is pressed by the locking cam **31** to be in an ON state, the locking cam **31** enters an initial state. When the synchronous motor **33** is driven from this state, the locking cam **31** collides with the stopper pin **34**A, and the rotation direction of the synchronous motor **33** is reversed from CCW to clockwise (CW). When the locking cam **31** rotates in the CW direction, the cam phase detection switch **32**A enters an OFF state, and the synchronous motor **33** is driven for a predetermined time period from that point in time. As a result, the locking cam **31** can be stopped at a position of the locked state (initial position) (the state illustrated in FIG. **12**A).
(74) To release the door locked state, when the synchronous motor **33** is driven, the locking cam **31** collides with the stopper pin **34**A and then rotates in the counterclockwise direction. As illustrated in FIG. **12**B, when the control unit **40** detects that the cam detection switch **32**A has been changed from OFF to ON, the driving of the synchronous motor **33** is stopped. As a result, the locking cam **31** stops at the rotational position corresponding to the door unlocked state.
Second Modified Example of First Embodiment
(75) A configuration in which the locking unit **30** of the above-described first embodiment is replaced with a locking unit **30**B will be described below as a second modified example. In this locking unit **30**B, the locking cam **31**B can be rotated in both directions by a stepper motor.
(76) FIG. **13**A is a plan view illustrating a rotational position of the locking cam **31**B in the door locked state of the locking unit **30**B according to the second modified example of the first embodiment of the present disclosure. FIG. **13**B is a plan view illustrating a rotational position of the locking cam **31**B in the door unlocked state of the locking unit **30**B. Note that in these drawings, to make it easier to see the rotational positions of the locking cam **31**B, illustration of the

stepper motor and the like constituting the locking unit **30**B is omitted.
(77) The locking unit **30**B includes the locking cam **31**B, a stepper motor (not illustrated), a stopper pin **34**B, and a pinion gear **35**B.
(78) This locking cam **31**B is provided with a cam inner groove **31**Ba and a cam outer peripheral gear **31**Bb. The stopper pin **34**B is disposed inside the cam inner groove **31**Ba to limit a rotation range of the locking cam **31**B to about 90 degrees. The pinion gear **35**B transmits rotation from the stepper motor to the cam outer peripheral gear **31**Bb to rotate the locking cam **31**B.
(79) In the locking unit **30**B, the stepper motor is driven by the motor drive circuit **42** to rotate the locking cam **31**B in the clockwise (CW) direction. Due to this rotation, one end of the cam inner groove **31**Ba of the locking cam **31**B comes into contact with the stopper pin **34**B, as illustrated in FIG. **13**A. Since the locking cam **31**B cannot be rotated any further and the stepper motor steps out, the driving of the stepper motor is stopped at this rotational position. As a result, the locking cam **31**B stops at an initial position corresponding to the door locked state.
(80) To release the door locked state, the stepper motor is driven in the reverse direction by a predetermined number of pulses to rotate the locking cam **31**B in the counterclockwise direction. By this rotation, the cam inner groove **31**Ba of the locking cam **31**B can be stopped at the rotational position corresponding to the door unlocked state without colliding with the stopper pin **34**B, as illustrated in FIG. **13**B.
(81) FIG. **14** is a perspective view of the locking cam **31**B as viewed from below.
(82) As illustrated in FIG. **14**, a groove **36**Bc for a flat head screwdriver is provided at a shaft portion of the locking cam **31**B, the shaft portion being provided on the lower side of the locking cam **31**B.
(83) Then, the locking unit **31**B provided with the above-described locking cam **30**B is disposed at or near the door opening button **15** inside the main body, and a hole into which a flat head screwdriver can be inserted is formed in the bottom surface of the main body. It is preferable that the hole be normally closed with a cap or the like.
(84) According to such a configuration, even if the door cannot be unlocked by operating the door unlock key **16***b* due to power failure or malfunction, a flat head screwdriver can be inserted to rotate the locking cam **31**B. As a result, the door **12** can be manually put in the door unlocked state and performing the pressing operation on the door opening button **15**.
(85) Embodiments of the present disclosure have been described above with reference to the drawings. However, the present disclosure is not limited to the embodiments described above, and the present disclosure can be implemented in various modes without departing from the gist thereof. Further, the present disclosure can be made in various forms by appropriately combining a plurality of components disclosed in the embodiments described above. For example, some components may be removed from all of the components described in the embodiments. For easier understanding, the drawings schematically illustrate respective main components, but the number of illustrated components or the like may differ from the actual number of components for the sake of convenience in creating the drawings. In addition, each of the components described in the above embodiments is exemplary and is not particularly limited, and various modifications can be made within a range that does not substantially depart from the effects of the present disclosure.
INDUSTRIAL APPLICABILITY
(86) The present disclosure is applicable to a field such as that of a heating cooking apparatus having a configuration in which a door of a front face opening of a heating chamber is opened by a mechanical operation.
(87) While preferred embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

## Claims

1. A heating cooking apparatus comprising: a main body including a heating chamber accommodating an object to be heated; a door configured to open and close a front face opening of the heating chamber; a locking member provided at the door; a holding member provided at the main body and configured to hold the locking member to cause the door to be in a locked state; a first operation member configured to be displaced between a first position and a second position and to be operated to open the door; a second operation member configured to be operated to release the locked state of the door; a displacement transmission mechanism configured to transmit a displacement of the first operation member from the first position, and to release the locked state of the door by the holding member by causing the first operation member to be displaced to the second position; a displacement transmission limiting member configured to be movable between a limiting position at which at least one of the displacement of the first operation member and the transmission by the displacement transmission mechanism is limited, and a non-limiting position at which neither the displacement of the first operation member nor the transmission by the displacement transmission mechanism is limited; and a driving unit configured to cause the displacement transmission limiting member to move to the non-limiting position in response to an operation of the second operation member.
2. The heating cooking apparatus according to claim 1, further comprising a control unit configured to electrically control the movement of the displacement transmission limiting member to the non-limiting position by the driving unit.
3. The heating cooking apparatus according to claim 1, further comprising a display unit configured to visually display whether the door is in the locked state.
4. The heating cooking apparatus according to claim 1, further comprising a notification unit configured to audibly notify that the door has entered the locked state or that the locked state is released.
5. The heating cooking apparatus according to claim 2, wherein the control unit causes the driving unit to move the displacement transmission limiting member to the limiting position before a heating operation is started.
6. The heating cooking apparatus according to claim 1, wherein the displacement transmission limiting member is configured to be manually movable to the non-limiting position from outside the main body.
7. The heating cooking apparatus according to claim 1, further comprising an operation unit provided at a front face of the main body and at which the first operation member and the second operation member are disposed, wherein the first operation member and the second operation member are separated from each other by one half or more of a length of the operation unit in a longitudinal direction of the operation unit.